Response to Reviewers' Comments

Paper: Why AI-Generated Image Detectors Fail on Modern Diffusion Generators: A Diagnostic Study of Cross-Architecture Generalization and Failure Modes
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All section, table and figure numbers below refer to the revised manuscript.

We thank the three reviewers for comments that were specific enough to act on directly. Every point is answered below with the location of the corresponding change. Four of the suggestions required new experiments, all of which we ran: a test of the ensembling hypothesis (Section 4.4), per-detector reliability diagrams (Figure 1), a matched-compression condition that quantifies the PNG/JPEG confound (Section 6), and a comparison against published numbers for the same checkpoints (Section 3.5). One suggestion, adding a further contemporary detector, we were only able to address in part, and we say so plainly below rather than claiming otherwise.

Reviewer 1

1.1 Abstract is compacted to highlight the main findings.
The abstract has been rewritten. It now opens by stating that this is a diagnostic study, gives the per-generator sample sizes, and reports the new ensembling and matched-compression results, while the preprocessing and compression cautions are stated as bounds on the absolute numbers rather than as closing remarks.

1.2 The systematic analysis of failure profiles and calibration under distribution shift is a major contribution.
Thank you. We have strengthened this part rather than leaving it as it was: Section 4.7 now carries per-detector reliability diagrams (Figure 1, also requested by Reviewer 3), which show that for every failing cell the largest confidence bin is the rightmost one. That makes the point visible in a way the scalar ECE in Table 4 cannot: the curves fall vertically rather than shifting toward lower confidence.

1.3 Training data finding is based on two detectors and should be appropriately qualified.
Agreed, and we have done two things. First, the closing paragraph of Section 4.3 now marks the claim as a hypothesis rather than a finding and states the specific weaknesses: it rests on one controlled pair, $n = 5$ generators, the permutation p of 0.017 is the floor of what the design can produce rather than an effect size, and a shared quirk of the SDv1.4 training subset could produce the same ordering twice for unrelated reasons. The abstract and conclusion were softened to match. Second, rather than only qualifying it, we tested a falsifiable prediction it makes (Section 4.4); the hypothesis survived that test, which is better evidence than restating it more cautiously would have been.

1.4 The leakage verification and McNemar tests are solid and should be kept as examples of evaluation practice.
Both are retained in full (Sections 3.4 and 4.2, Table 5). The McNemar machinery is also now reused for the new ensembling comparison.

1.5 Including at least one contemporary detector for diffusion models would widen the comparison.
Addressed in part, and we want to be accurate about which part. The evaluated set already contains two detectors built for the diffusion era, NPR (CVPR 2024) and the OMAT-hardened CLIP-LoRA (2025), the latter designed and published specifically against modern diffusion and Diffusion Transformer outputs; Table 1 now carries a Year column so this span is visible rather than having to be inferred. What we could

not do within the revision window is add a detector from a structurally independent family, meaning one whose detection signal is neither a CLIP embedding nor a pixel residual. The obvious candidate is a diffusion-reconstruction method such as DIRE, which requires running a diffusion model's inversion and reconstruction over every evaluation image; that does not fit the inference-only, single 8 GB GPU setting this study was built under and would have meant reporting a partial sweep. Section 8 states this as a limitation and names it as the most valuable single addition future work could make, rather than leaving the reader to discover the gap.

1.6 Adding a short list of practical recommendations for practitioners would be useful.

Added as Section 7: six recommendations for selecting and deploying a detector, each tied to a specific measurement in the paper, plus one recommendation addressed to benchmark authors regarding scriptable verification subsets.

Reviewer 2

2.1 Confounding due to JPEG vs PNG compression is an important note that could be highlighted earlier in the abstract.

Moved. The compression confound is now stated in the abstract with its measured size, and it appears in the introduction's contribution list rather than only at the end of the paper. It is also no longer a subsection buried inside a robustness section: Section 6 treats the robustness question and the benchmark confound together, because the single experiment answers both.

2.2 The analysis of resize conventions is a practical contribution that deserves to be highlighted.

Promoted. It is now one of the five numbered contributions in the introduction, has its own top-level section (Section 5) rather than sitting under a discussion heading, and generates recommendation 4 in Section 7. The result that two checkpoints trained on identical data prefer opposite preprocessing is stated in the abstract.

2.3 The manuscript should clearly state its nature as a diagnostic study.

Done in three places. The title now reads "A Diagnostic Study of Cross-Architecture Generalization and Failure Modes"; the abstract's first sentence says the study trains nothing and proposes no new detector; and the introduction has a short paragraph making the same point before the contribution list.

2.4 Being over 8,000 words long, the manuscript is too long for a conference and should be shortened.

Material carried over from the submitted version has been cut by roughly a fifth. Because the revision also absorbs four new experiments, three new tables and a new figure, all requested by the reviewers, the net length is close to that of the submitted manuscript rather than below it, and we would rather be explicit about that trade than claim a reduction the page count does not show. The cuts came from merging the three preprocessing subsections into one section with a single subsection; compressing the spectral-deviation analysis from roughly a page to one paragraph and removing its figure, since it is a negative result that did not need the space; folding the "generator easiness is detector-relative" subsection into Section 4.1; removing the per-image score histogram, whose evidence the new reliability diagrams carry more rigorously; and tightening the related work, methodology, limitations, ethics and conclusion sections throughout. No result was dropped. Should the committee need a further reduction, we are happy to cut the spectral-deviation subsection and the published-number comparison table, which would remove roughly another page. This response is submitted alongside the manuscript and is not part of the manuscript length.

2.5 Results for matched compression in real and synthetic cases would give a quantitative estimate of the confound effect.

This was the most useful suggestion we received and we ran the experiment. Section 6 and Table 8 now report three conditions on one fixed FLUX.1 sample of 2,000 images per class, paired within each detector: the benchmark as shipped; a *fakes-only* condition that re-encodes generated images at JPEG q75 and

leaves real images untouched, which removes the format asymmetry and nothing else; and a condition re-encoding everything. The result is sharper than we expected. Removing only the format asymmetry takes NPR from AUC 97.10 to 39.13, which is below chance, meaning the detector then ranks real photographs as more likely generated than the generated images. We note in the text why that matters for interpretation: destroyed evidence would drive AUC toward 50 and stop, so an inversion indicates the quantity NPR reads tracks relative compression level rather than provenance. The same manipulation costs the CLIP baseline 4.5 AUC points, the ResNet-50 4.3, CLIP-LoRA 12.8 and UniversalFakeDetect 1.0, against 58.0 for NPR, so the confound inflates an artifact-based detector by an order of magnitude more than a semantics-based one. For comparison, Grommelt et al. report shifts above 11 points when removing JPEG and dimension confounds from GenImage.

2.6 The ensembling hypothesis could be tested in a small experiment to support the conclusion.
Done, as a new Section 4.4 with Table 6. The submitted version predicted that combining detectors chosen for different training lineages should beat combining for architectural variety; the per-image scores already collected let us test that without retraining. Holding the GenImage++ CLIP baseline fixed and varying only its partner, a same-lineage partner of different architecture takes worst-generator fake-accuracy from 9.6% down to 0.3%, while a different-lineage partner lifts it to 62.9%. A paired McNemar test separates the two combinations on all five generators ($p < 10^{-200}$ throughout). Across all pairs, excluding the nested CLIP/CLIP-LoRA pair, cross-lineage pairs average 52.5% worst-generator accuracy against 37.3% for same-lineage pairs. We also report two findings that cut against a simple "ensemble more" message and would have been easy to omit: lineage diversity is necessary but not sufficient, since a cross-lineage pair with a weak partner underperforms the strong detector alone, and every pair we construct is beaten by the single best detector. The conclusion is phrased accordingly, as a statement about where blind spots come from rather than a leaderboard claim.

Reviewer 3

3.1 Abstract should mention the sample sizes for each generator.
Added: 6,000 real and 6,000 generated images for BigGAN, GLIDE, FLUX.1 and SD3, 18,300 per class for SDXL, and 2,000 per class for the paired compression experiments.

3.2 Sharing the evaluation code and preprocessing parameters would improve the significance of the work.
We agree, particularly given that Section 5 argues precisely this point about other people's releases. Appendix A now itemizes what is released rather than promising a repository in general terms: the harness and the exact command line for each of the 25 detector/generator cells, the per-image prediction scores every table and test in the paper is computed from, the preprocessing functions for the three resize conditions together with the normalization statistics used per detector, and the scripts that regenerate every results table and the figure from the score files. The repository is public at `https://github.com/MogSquad-FinalYear/BlindSpot`.

3.3 Calibration analysis could be extended by adding reliability diagrams for each detector.
Added as Figure 1: one panel per detector, five generators each, plotted in the same coordinates the reported ECE integrates so the figure and Table 4 are consistent by construction. Marker area is proportional to bin occupancy, which we found necessary because almost all mass sits at confidence near 1 and an unweighted curve makes sparse middle bins look as informative as the extremes; bins with fewer than 25 images are omitted. The final paragraph of Section 4.7 reports what the diagrams add beyond the scalar.

3.4 Choosing the five generators should be motivated in terms of covering current generator families.
Section 3.2 now opens with that motivation. The set covers the three denoiser and generator families that account for essentially all current open-weight synthesis, with the U-Net family given two mem-

bers (GLIDE and SDXL) so it contributes a within-family contrast of four years rather than a single point. We also name the two families the study does not cover, autoregressive token-based generators and face-specific manipulation pipelines, the second deliberately.

3.5 Comparisons with results from GenImage and other benchmarks would help the reader to relate these findings.

Added as Section 3.5 and Table 2, placed in the methodology so the harness is validated before new numbers are reported. We compare our overall accuracy against the figures the NPR repository publishes for the same ProGAN-4class checkpoint we evaluate: three of four cells agree within five points and the UniversalFakeDetect GLIDE cell agrees to within 0.1. We flag that those subsets are assembled differently from GenImage's, so this is a consistency check rather than a reproduction. The one same-benchmark comparison available, against Zhou et al.'s reported 62.66% for their CLIP baseline on BigGAN, is the disagreement that Section 5 resolves, and Section 3.5 now forward-references it instead of leaving the reader to encounter the 6x gap unprepared.